

Annexure 2

**Evidence of Course requirement from industry
(The job/ Skill requirements are highlighted in the following news/ads)**

A quick guide to a career in Cybersecurity: Qualifications, skills, job description and salary

Cybersecurity has become one of the most in-demand careers in India now as governments and organizations are facing cyber threats. Here are the qualifications needed, skills you must have and the salary you will get for a career in cybersecurity.



[India Today Web Desk](#)

New Delhi

November 13, 2019

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Here are the qualifications needed, skills you must have and the salary you will get for a career in cyber security.

Governments and organizations across the globe are increasingly vulnerable to cyber threats. According to [research](#), there is an attack every 39 seconds on the web. Well, sites are getting hacked every day, and the world is in need of quality cyber security experts.

There is a growing demand for people to be trained in internet security to help defend against cybercrimes. As a student or a professional, if you want to make a career in Cyber security, this is a great opportunity.

In this quick guide, we will explore the different aspects required to start a career in Cyber security. Let's get started.



Education qualifications you need for a cybersecurity career

In order to start a career in Cybersecurity, you need to start off with a background in Information technology.

While a bachelor's degree focused on information and technology is one of the most prerequisites, you must have in-depth knowledge of the latest industry trends.

Getting a bachelor's degree in Computer Science is the best way to leap towards the cybersecurity domain.

The following are your options to pursue after 12th to embrace a career in Cybersecurity:

- B.Tech (Information Technology)
- B.Tech (Information Science & Engineering)
- B.E. (Information Technology)
- Diploma in Cyber Security
- B.Tech (Cyber Security and Forensics)

Aspiring candidates may have to take entrance exams like JEE (Joint Entrance Exam) to take admission to these courses.

If you have completed your bachelor's degree, you can look for a job in numerous organisations as a cybersecurity expert. However, if you want to pursue this field more through academics, you can go for Master's degree course in Cybersecurity.

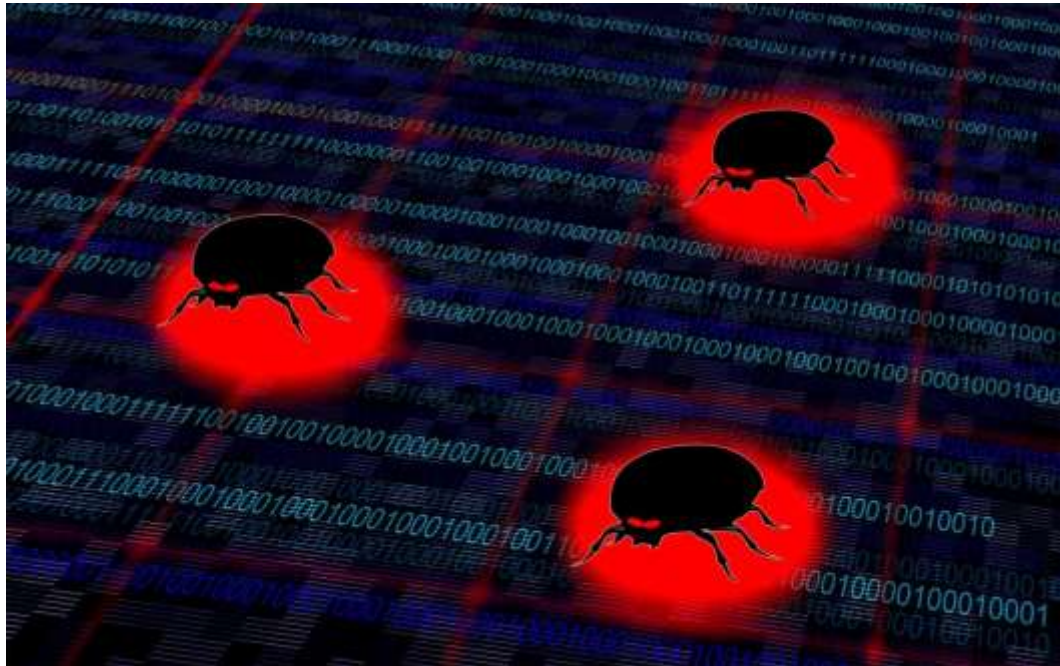
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The courses may include:

- Master of Computer Application (MCA)
- M.B.A. (Information Security Management)

- M.E. (Information Technology)
- M.Tech (Information Technology)
- M.Tech (Cybersecurity)
- Post Graduate Diploma in Cybersecurity

Within the realm of engineering and computer science degrees, you will find specializations that will help you land different cybersecurity roles.



Skills needed for a career in cybersecurity

However, being a cybersecurity expert, you will need to possess specific skills that will help you tackle problems with ease. This may include:

- Strong ability to problem-solving since you have to do this at every minute of your job.
- You must have experience with an extensive range of security tools and computer systems for executing programs in an effective manner.
- You must have strong analytical skills with proven capability to communicate with a variety of people within your company as well as partner organisations.
- You must have the ability and flexibility to multi-task in a fast-paced, tense atmosphere.
- As a cybersecurity expert, integrity and dependability is the last thing required since you have to safeguard the interests of the company you are working for.

What does a Cybersecurity expert do?

- As a cybersecurity expert, your job will include doing everything from safeguarding the network to implementing information security programs, and coordinating with other industry experts and implementing a plan.
- You will diagnose and monitor viruses, malware, and other vulnerability issues and make recommendations for solutions. This includes both software and hardware programs.
- You will be responsible for designing firewalls, regulate access, and monitor the use of data files to protect the network and safeguard information.
- One of your duties include staying aware of the latest malware and viruses in-order to inhibit these viruses from entering your company's network.

- You will also be responsible for promoting security awareness, imparting training to the end-users, provide updates and reports to management, and develop policies about numerous aspects of Cybersecurity.



How much salary does a cybersecurity expert get?

In India, the average salary for a cybersecurity expert is Rs 79,846 per month.

However, it depends on the location and the organisation you are working for.

With the growing demand for cybersecurity specialists in India and the rest of the world, it can be said that the pay grade will be higher than expected if you have the required knowledge and experience.

- Article by Gaurav Vohra, Co-Founder and CEO, Jigsaw Academy

Job Outlook For Cyber Security Professionals



It's been estimated that cyber crime could cost the world over \$6 trillion annually by 2021. Fears surrounding hackers have been underscored by disruptions in political processes, social media accounts, and more. This has helped set in motion a record number of job openings for information security professionals.

There are a number of different industries that are especially in need of this kind of expertise. Apart from the political sphere, banks and financial institutions have been experiencing a more recent uptick in cyber attacks.

According to a [2016 survey](#), banks experience an average of 85 serious attempted data breaches every year. 36% of those attempts result in hackers successfully obtaining some sensitive information.

The information-intensive healthcare industry has also expressed a need to increase their information security capabilities following the 2017 WannaCry ransomware attack, a stunt which devastated national health services around the globe.

By the time the attack was over, a quarter-million machines in more than 150 countries had been affected. The NHS, Spain-based Telefonica, America's FedEx, German railway company Deutsche Bahn, and LATAM Airlines were all impacted.

Thanks to their wealth of intellectual property, campus research, and personal records, educational facilities are seeing increased incidents of cyber attacks as well. The University of California, San Francisco recently experienced a cyber attack just a few months ago.

On June 1, cybercriminals used NetWalker malware to encrypt data on the servers of the university's school of medicine. Students and faculty alike were told they would be unable to access their devices until a ransom in cryptocurrency was paid.

In total, the university was forced to hand over \$1.14M to recover its encrypted data. With that, it's safe to say those hoping to land a cybersecurity position will be met with a fair amount of job opportunities.

Indeed, cybersecurity professionals are in high demand. These individuals have the option of specializing in certain areas including information technology or IT, ethical hacking, information defense, information assurance, defense engineers, risk managers, network protection, and more.

Types of Cyber Security Job Titles



The cybersecurity workforce is multifaceted and made up of individuals specializing in unique areas. Check out the list below to explore different titles associated with the field.

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Security Specialist

Many individuals interested in a cybersecurity career first look towards becoming a security specialist. This entry-level position requires a bachelor degree.

An information security specialist is expected to protect software and network security systems. They must keep an eye on defense updates and improvements, monitor administrations, ensure all tools are running properly, and develop system specialized security requirements.

The national average salary for a Security Specialist is \$111,052 in the United States.

Incident Responder

Incident responders respond to information hacks and emergencies. They are typically the first ones on the scene of cyberattacks or when an organization's network or system has been compromised.

They are also responsible for helping prevent any major attacks from happening again. They must also be able to recognize existing vulnerabilities in a network or system, oversee systems and applications for suspicious activity, run penetration tests, risk analysis, and audits.

These individuals should be able to develop a system to spread awareness during an emergency and be able to relay all necessary information to law enforcement. Well-composed incident reports should also be created and handed off to upper management.

A Cyber Incident Responder will typically earn around \$110,972 annually.

Security Administrator

These individuals are responsible for installing and monitoring security controls.

Security administrators must take certain security measures, including supervising systems for suspicious activity or traffic, running audits and providing policy recommendations, protecting computer systems against illegal access, and creating plans for recovery should a breach occur.

The average annual pay for a Cyber Security Administrator in the United States is \$91,661 a year.

Vulnerability Assessor

Also on the list of cybersecurity jobs are vulnerability assessors. These individuals spend their days searching and analyzing possible errors in computer networks and information systems. They also provide recommendations to businesses on how to improve their network security systems.

They may also be asked to compose and describe vulnerability assessments, test vulnerabilities by creating and testing custom scripts and applications, and use creative strategies to highlight additional risks.

The median salary for a Security Assessor is \$90,000.

Cryptographer

Job alerts may also point recent graduates towards cryptography. Individuals working in this area create ciphers, algorithms, and security systems using code. These individuals must make sure their scripts are able to protect data from unauthorized users, create systems to guard against exposures, and make sure critical information is protected from being edited, copied, or deleted.

They are also responsible for testing systems for vulnerabilities, analyze data to solve any existing safety issues and keep up-to-date with current research and strategies for coding.

Cryptographers earn average salaries of just over \$73,000.

Security Manager

These individuals fall more into the managerial sphere than some of the more technical jobs listed above. These individuals will be responsible for assembling a reliable team of cyber security experts, manage budgets, assess new software, and remain up-to-date with industry trends.

The national average salary for a Cyber Security Manager is \$51,881 in the United States.

Security Manager

These individuals design systems, manage employees, and assess the needs of their business or organization. They must also test their systems for any vulnerabilities and implement employee protocols to maintain system integrity.

Salaries for a Cyber Security Manager can reach up to \$138,690 in certain parts of the United States.

Chief Information Security Officer (CISO)

Largely considered a more advanced position within the field, a Chief Information Security Officer must oversee all defense policies and procedures for their company. Some organizations may require applicants to possess a masters in cybersecurity.

They must remain current on all industry trends, including how cyber criminals are behaving. They must also remain aware of how all company decisions are made and provide input from a security perspective.

The 2020 average Chief Information Security Officer salary in the US is \$221,601.

Information Security Analyst

Information security analysts must analyze all policies and protocols for their company. They must also conduct thorough audits to determine any existing weaknesses in the company's cyber defense system. They must also anticipate future developments in digital security crime and work accordingly to prevent potential attacks.

The average salary for information security analysts falls between \$90,000 and \$160,000.

Security Auditor

These network security professionals work with risk management must produce detailed reports on system performance. They must also make sure all records are protected by firewalls, encryption, and other cyber defense measures.

The average salary for this position falls at around \$66,000.

Forensic Expert

These individuals must examine all security breaches, provide perspective on how it took place, and take measures to locate the criminals involved. They must also train first responders on how to handle evidence including computers, hard drives, or portable drives.

The average annual pay for a Computer Forensics Analyst in the United States is \$100,063 a year.

Source Code Auditor

Individuals holding this title must review all source code in a company's systems or applications to find any security vulnerabilities or defects that might affect its ability to protect against security threats. This means they must be aware of any technologies or trends popular among cyber criminals. The position demands an in-depth understanding of computer science.

Cybersecurity Talent Crunch To Create 3.5 Million Unfilled Jobs Globally By 2021

350 percent growth in open cybersecurity positions from 2013 to 2021

The 2019/2020 Official Annual Cybersecurity Jobs Report is sponsored by [Herjavec Group](#), a leading global cybersecurity advisory firm and Managed Security Services Provider (MSSP) with offices across the United States, Canada, and the United Kingdom.



– [Steve Morgan](#), Editor-in-Chief

Sausalito, Calif. – Oct. 24, 2019

The New York Times [reports](#) that a stunning statistic is reverberating in cybersecurity: Cybersecurity Ventures' prediction that there will be 3.5 million unfilled cybersecurity jobs globally by 2021, up from one million positions in 2014.

The cyber employment figure has been corroborated by hundreds of media outlets, including the world's largest, as well as industry associations, universities, governments, vendors, recruitment firms, and security experts, since our original report was published in [May 2017](#).

Soon after our employment data was released, the World Economic Forum (WEF) republished an article with permission from Knowledge@Wharton, the online research and business analysis journal of the Wharton School of the University of Pennsylvania, which shared our report, and observed [“nowhere is the workforce-skills gap more pronounced than in cybersecurity.”](#)

Earlier this year, the Harvard Business Review shared our report, and summed up the plight: “The majority of chief information security officers around the world are worried about the cybersecurity skills gap, with 58 percent of CISOs believing [the problem of not having an expert cyber staff will worsen.](#)”

Cybersecurity Ventures arrived at our original estimation of unfilled positions after reviewing and synthesizing dozens of employment figures from the media, analysts, job boards, vendors, governments, and organizations globally. In 2019, we've conducted similar research, and we stand firmly behind the two-and-a-half-year-old prediction.

Over the eight-year period tracked, the number of unfilled cybersecurity jobs is expected to grow by 350 percent, from one million positions in 2013 to 3.5 million in 2021. And of the candidates who are applying for these positions, [fewer than one in four are even qualified](#), according to the MIT Technology Review.

Despite industry-wide efforts to reduce the skills gap, the open cybersecurity headcount in 2021 will be enough to fill 50 NFL stadiums. A Forbes Technology Council [post](#) states that the number would be in the top 100 for the size of work populations of countries worldwide — not just industries.

Every IT position is also a cybersecurity position now. Every IT worker, every technology worker, needs to be involved with protecting and defending apps, data, devices, infrastructure, and people.

[“It’s a full-on war for cybertalent”](#) one industry veteran who specializes in recruiting and placing senior-level information security executives told the Los Angeles Times.

Robert Herjavec, founder and CEO at [Herjavec Group](#), said, “Unfortunately the pipeline of security talent isn’t where it needs to be to help curb the cybercrime epidemic. Until we can rectify the quality of education and training that our new cyber experts receive, we will continue to be outpaced by the Black Hats.”

“There is a [zero-percent unemployment rate in cybersecurity](#) and the opportunities in this field are endless,” added Herjavec. “Gone are the days of siloed IT and security teams. All IT professionals need to know security – full stop. Given the complexity of today’s interconnected world, we all have to work together to support the protection of the enterprise.”

Numerous pundits from industry associations and vendors have chimed in on the cybersecurity jobs dilemma, with adjustments to shortfalls in their previous estimates, which now align more closely with Cybersecurity Ventures’ steady forecast.

ISC(2), an international, nonprofit membership association for information security leaders, produced research in 2018 which estimated the cyber worker shortage at nearly [3 million](#) — on the heels of a report they put out just a year earlier, which stated a drastically lower [1.8 million](#) figure by 2022. Members of the ISC(2) leadership team recently posted a [blog](#) on their website which states “the cybersecurity skills shortage is expected to result in 3.5 million unfilled positions by 2021” (according to Cybersecurity Ventures via Forbes) — which underscores ISC(2)’s alignment to our research. “By Cybersecurity Ventures’ predictions, and we agree from everything we see, there will be [3.5 million unfilled cybersecurity jobs in the next couple of years worldwide](#),” said Mark Aiello, a cybersecurity recruiting expert and representative for the eastern Massachusetts chapter of ISC(2) at a recent cybersecurity conference held in Boston.

Our estimate has solidified as the de facto figure which the cybersecurity industry and the media relies upon for projected job openings.

WIRED highlights our jobs data in a recent story and concludes that despite looming threats of an ever smarter and technologically advanced hacking landscape, the [security industry is not resourced to fight hackers or AI-powered exploits](#).

A Forbes Business Development Council [post](#) asserts that if you aren't prioritizing hiring or training for IT security in 2019, then it's time to rethink your company's core priorities. If you agree, then read on...

The editors at Cybersecurity Ventures have compiled the most important facts, figures, statistics, and predictions to help frame the global cybersecurity labor shortage, and what is being done in an attempt to close the gap.



UNFILLED JOBS

No matter how you crunch the numbers, there's a huge need for cybersecurity workers over the next decade.

- Out of the 3.5 million open cybersecurity positions expected by 2021, Cybersecurity Ventures estimates more than 2 million openings will be in the Asia-Pac region, and nearly 400,000 will be in Europe.
- The U.S. has a total employed cybersecurity workforce consisting of 715,000 people, and there are currently 314,000 unfilled positions, according to [Cyber Seek](#), a project supported by the National Initiative for Cybersecurity Education (NICE), a program of the National Institute of Standards and Technology (NIST) in the U.S. Department of Commerce.
- Members of the CNBC Tech Council put the number of U.S. technology job openings at [700,000 to 1 million](#) over the past 12 to 24 months. Based on those figures, cybersecurity positions made up 32-45 percent of all U.S. tech job openings.
- At the end of 2018, there were more than [26,000 openings for cybersecurity analysts in the U.S.](#)
- Cybersecurity isn't alone in the labor crunch, although it's one of the hardest hit STEM fields. Between 2017 and 2027, [STEM jobs will grow 13 percent](#), compared to 9 percent for all other jobs. However, it was estimated that in 2018, [4 million STEM jobs in the U.S. went unfilled](#).

There will be 3.5 million unfilled cybersecurity jobs by 2021, up from 1 million positions in 2014.



U.S. METROS

There are many new initiatives across the U.S. that are aimed at bolstering our nation's future pipeline of cybersecurity candidates.

- Government Technology recently reported that Massachusetts has more than 9,000 open cybersecurity jobs. The MassCyberCenter is looking to boost the state's cyber resilience and economic growth and to partner with businesses, academia and the public sector to [train new cybersecurity workers](#).
- To develop the cyber workforce of the future, the New York City Economic Development Corporation (NYCEDC) [launched Cyber NYC](#), which is supposed to catalyze 10,000 cyber jobs, help fuel new startups, and protect the city's anchor employers. The project is among the [nation's most ambitious cybersecurity initiatives](#), which over the next decade could transform New York City into a global leader of cybersecurity innovation and job creation.
- The [New York metro area](#) (NY-NJ-CT) has the second-largest absolute number of cybersecurity job openings – roughly 20,000.
- Employers in the Philadelphia area struggled to fill more than [10,000 open roles requiring cybersecurity skills](#) last year. In [Philadelphia](#), the biggest skill shortages fall within the following areas: security analysis and investigation, application security, and cloud computing security.
- The population of cyber engineers and analysts throughout the [Washington D.C. Beltway](#) is 3.5 times as big as the rest of the U.S. combined.
- With more than 150,000 cyber-related engineering and data science professionals, [Maryland has the number one cyber workforce in the world](#), and leads the U.S. in cyber employment for classified nation-state jobs. Maryland also has the largest concentration of university-trained cyber engineering graduates in the world.
- There are currently over [15,000 open cybersecurity jobs in Maryland](#) — and that number is growing by the day, according to the Maryland Chamber of Commerce.
- The Northern Virginia Technology Council noted that employers in the region are struggling to find candidates who have both the technical and soft skill competencies they need. Employers cite [soft skills deficiencies](#) as their primary pain point outside of the obvious shortage of qualified candidates.
- The [top five states for cybersecurity jobs](#) are Virginia, Texas, Colorado, New York and North Carolina — where average annual salaries for cybersecurity roles exceed \$100,000, according to DarkReading's analysis of a recent report.
- The Georgia Cyber Center, a \$100 million facility, the country's largest cyber investment of its kind by a state government, is intended to [train the IT security workforce of the future](#).
- The University System of Georgia signed an agreement with the U.S. Army Cyber Center that will allow active duty and reserve military members to work toward a degree in cyber fields at several of

Georgia's universities during their service. The Army intends for the program to help [fill over 11,000 open cyber jobs in the state](#), both inside and outside the armed forces.

- The Cyber Innovation Center in [Bossier City, Louisiana](#), which has new Homeland Security funding, expects to broaden its cyber skills preparation to 10 million students and 50,000 teachers in K-12 across the U.S. — which should eventually bolster the pipeline of young cyber workers.
- [Tech employment in Chicago](#) grew by nearly 6,000 jobs last year, and cybersecurity analyst is one of the top 3 tech positions in the windy city.
- [San Antonio](#) is home to the nation's second-largest concentration of cybersecurity experts. With a pipeline of cybersecurity talent from San Antonio's K-12, up through top higher education programs and advanced military backgrounds, the city, a.k.a. [Cyber City, USA](#), is preparing for jobs employers need to fill today and in the future.
- [Texas](#) has the most cybersecurity growth potential in the U.S., according to Business Facilities' 15th Annual Rankings Report.
- [Arizona](#), which the Federal Trade Commission ranked as one of the top 10 in cybersecurity, has roughly 7,000 cybersecurity job openings.
- San Diego, home to the U.S. Space and Naval Warfare Systems Command (SPAWAR), and home to more than 150 cybersecurity companies, has around 8,450 cybersecurity jobs, which is [up by 11 percent from three years ago](#), according to the region's Cyber Center of Excellence.
- California has 50 companies (33 percent) named on Cybersecurity Ventures' [Hot 150 Cybersecurity Companies To Watch In 2020](#) list, the bulk of whom are located in or in close proximity to Silicon Valley. These plus hundreds of others in this region have created an extraordinarily competitive market for sales, marketing, and engineering talent.



INTERNATIONAL

Help Wanted: Cybersecurity experience preferred. This is the consistent theme across regions globally.

- India — by far — had the [largest growth in cybersecurity job openings](#) from 2017 to 2018, followed by Sweden, Ireland, Italy and the UK.
- The National Association of Software and Services Companies (NASSCOM) estimates that India — a country with a population of approximately 1.34 billion people — alone will need [1 million cybersecurity professionals by 2020](#) to meet the demands of its rapidly growing economy.
- Skill shortages in Europe are damaging the growth prospects of companies and the continent's economy, according to an EY survey. The most [alarming talent shortages](#) are in digital skills, with cybersecurity (48 percent) identified as one of the scarcest skills.
- A recent parliamentary inquiry found [the cybersecurity worker shortage is “verging on a crisis” in the UK](#). The joint committee on the national security strategy's review found even the National Cyber Security Centre (NCSC), the government body charged with providing cybersecurity advice and support, faces a constant challenge to recruit the expertise it needs.
- [Ireland's skills gap continues to increase](#), particularly in the area of cybersecurity, where there's growing demand for roles year on year. The country is unable to keep up with its 18 percent increase in the demand for cybersecurity roles.

- Israel, the world's no. 2 exporter of cyber technology and one of the [top 10 global cybersecurity hubs](#) for 2019, is experiencing a [worker shortage](#) that is the most significant problem the sector is facing today. This shortage is causing local salaries to surge and is pushing firms to seek workers abroad, according to a recent [report](#) by Start-Up Nation Central and the Israel Innovation Authority.
- A report coming out of Australia found that 88 percent of IT decision-makers believe there is a [shortage of cybersecurity skills](#) within their own organization, but also nationally.
- To address its cyber skills shortage in South Africa, [Absa](#) has collaborated with the Maharishi Institute (MI) to set up a Cybersecurity Academy. The program is aimed at empowering marginalized youth, who would otherwise not have had access to tertiary education. [The learners who participate become certified cybersecurity analysts.](#)
- [Nigeria has a chronic shortage of cybersecurity workers.](#) With an estimated population close to 200 million, the number of certified cybersecurity professionals in Nigeria was a mere 1,800 as of last year. Microsoft has called for [safer online community](#) in Nigeria for improved economic growth.
- Research from Deloitte shows a lack of talent is being felt across corporate Canada. Critical roles are going unfilled, and it's expected that organizations across [Canada will need to fill an estimated 8,000 additional cybersecurity positions by 2021.](#)
- To help grow a security-trained workforce, the Cisco Networking Academy has partnered with 1,550 academies and 3,800 instructors to provide [cybersecurity training across Latin America](#). As of last year, some 1.6 million had already completed the coursework.



TRENDS

If there's one trend that everyone can agree on, it's that cybersecurity is a fast-growing market with tremendous career opportunities.

- The cybersecurity unemployment rate is at [zero percent](#) in 2019, where it's been [since 2011](#).
- Cybersecurity Ventures forecasts that 100 percent of large corporations (Fortune 500, Global 2000) globally [will have a CISO](#) or equivalent position by 2021 (up from 70 percent in 2018), although many of them will be unfilled due to a lack of experienced candidates.
- The U.S. Bureau of Labor and Statistics projects that employment for [information security analysts will grow at 32 percent between 2018 and 2028](#), which is higher than the average for all other occupations.
- Readers Digest named [cybersecurity project managers](#) and blockchain developers on its list of the 21 most in-demand jobs for 2020.
- Jobs requesting public cloud security skills remain open 79 days on average — longer than almost any other IT skills, according to Cyber Seek.
- A Forbes HR Council post observes that 74 percent of companies recently surveyed said that [the skills shortage is impacting their business](#), including the ability to keep their information secure.
- Tech executives on the CNBC Technology Executive Council said it has become harder to fill IT and cybersecurity positions, so candidates with liberal arts degrees, or [no college degree](#), are now being hired.

- CISSPs (Certified Information Systems Security Professionals) [rank sixth](#) out of the top 20 fastest-growing skills for freelancers.
- According to a DarkReading report published last year, only 14 percent of IT security managers felt there [were enough cybersecurity professionals](#) in the field with the needed skills to hunt down and respond to threats.
- The cybersecurity labor shortage is felt most acutely in the government workforce. Federal agencies, such as the Department of Defense — which houses both the National Security Agency and the U.S. Cyber Command — have [a hard time competing with the private sector](#) for top cyber talent.
- One of the [5 biggest trends in the job market](#) is a rise in cybercrime, which is spurring growth in security skills, according to CNBC.
- [Debby Carreau](#), CEO & founder of Inspired HR and a member of the CNBC-YPO Chief Executive Network, lists [cybersecurity](#) as one of five hot industries that are a safe bet for a career that's future-proof.
- Retention is a major issue for employers. According to a McAfee [survey](#) of 950 cybersecurity managers and professionals at organizations with 500 or more employees in the U.S., U.K., Germany, France, Singapore, Australia, and Japan, a full 89 percent of respondents would consider leaving their roles if offered the right type of incentive.



YOUTH MOVEMENT

We need to encourage and nurture our K-12 and college students to become future cyber fighters.

- One of the keys to building a future cybersecurity workforce is instilling enthusiasm for STEM and cybersecurity at an early age. The first-ever national [Cybersecurity badges for girls](#) in grades K–12 were announced by the Girl Scouts of the USA in 2017 and there has been strong industry support for the initiative.
- The National Security Agency has been [training kids to root out cybercriminals](#) since 2014. This past summer through a program called GenCyber, the N.S.A. ran 122 cybersecurity camps — named Camp Cryptobot — across the country, in order to generate a future pipeline of cyber workers.
- In one [report](#), across 24 states, only 35 percent of high schools in the U.S. teach computer science — much less focus specifically on cybersecurity — despite the fact that 90 percent of parents want computer science education for their children, [according to Ralph P. Sita](#), co-founder and CEO at Cybrary, a provider of free online cybersecurity training.
- According to McAfee's Winning the Game report, 92 percent of cybersecurity managers say [gamers possess skills that make them suited to a career in cybersecurity](#) — and 75 percent would consider hiring a gamer even if that person had no cybersecurity training or experience.
- By 2025, millennials will make up 75 percent of the global labor force — and more than 70 percent of the ethical hacker community is already [under the age of 30](#). Organizations therefore need to tap into this community in order to build their cybersecurity workforce.
- [Just 4.2 percent of federal cybersecurity workers are aged 30 and under](#). These people are charged with protecting the country's digital infrastructure via jobs in such departments and agencies as the

Department of Defense, CIA, Federal Election Commission and Department of Energy (which designs, tests and creates nuclear weapons for the U.S., among other tasks).

- [The Yale Tribune](#), a student run newspaper from Yale University, informs that cybercrime will cost the world [\\$6 trillion annually in 2021](#), double the cost from \$3 trillion in 2015, according to Cybersecurity Ventures. They advise that with strong coding skills, analysis, and certification, you can land a lucrative job in the cybersecurity industry — where demand outweighs supply.



COMPENSATION

Cybersecurity is a highly respected and well-paying occupation, from entry-level positions up to chief information security officers.

- The [top 10 percent of cybersecurity analysts make over \\$117,000](#) per year, while the median annual salary is \$76,000.
- New data indicates that of all IT jobs, cybersecurity engineers — with an average annual salary of \$140,000 — were the [highest paying and most recruited](#) heading into 2019.
- For the top coders with leadership and cybersecurity skills — a rare breed — [salaries exceed \\$225,000](#). In some companies, this position pays more than it does to the CISO. Software plus “soft skills” equals big pay for aspiring programmers with a senior management role in their sights.
- The [second-highest paying tech job in 2019](#) is a CISO, with a salary range of \$175,000 to \$275,000. Fortune 500 corporations in big cities pay as much as [\\$380,000](#) to [\\$420,000](#) annually, and more, to their CISOs, much higher than the average range for the position in mid-sized companies, government agencies, and academia.
- At the top of the cybersecurity food chain (ranked by pay), there are a small number of [CISOs earning 7-figure annual pay packages](#).
- Flaws in software code, which create vulnerabilities, have created a burgeoning bug bounty economy with big payouts to elite freelancer hackers. Some of them earn more [than \\$500,000 a year](#). But, that’s a far cry from the average take-home pay for most bug bounty hunters who are self-employed part-timers with no guaranteed income.

Cybersecurity engineers — at \$140,000 per year — were the highest paying and most recruited IT Jobs in 2019.



WOMEN IN CYBERSECURITY

Women know cyber, and we need more of them in the cybersecurity industry. Their talents and contributions are essential to the well being and safety of the world.

- Cybersecurity Ventures predicts that [women will represent 20 percent of the global cybersecurity workforce](#) by the end of 2019. This recalculates a 6-year old figure based on a limited survey that concluded women held just [11 percent](#) of cybersecurity positions.
- Research firm Forrester predicts that [the number of women CISOs at Fortune 500 companies will rise to 20 percent in 2019](#), compared with 13 percent in 2017. This is consistent with new research from Boardroom Insiders which states that [20 percent of Fortune 500 global CIOs are now women](#) — the largest percentage ever.
- Women in the cybersecurity field are [trending up in Israel](#), the world's second-largest country in terms of cybersecurity investment. In 2018, TechCrunch reported that for the most recent year tracked, 15 percent of newly established Israeli cybersecurity startups had a female founder, an increase from 5 percent the previous year.
- [91 percent of women in cybersecurity have a bachelor's degree](#), and 20-25 percent of them have an MBA or master's degree. 5 percent have a Ph.D., and 2 percent have no degree.
- A new book published by Cybersecurity Ventures — "[Women Know Cyber: 100 Fascinating Females Fighting Cybercrime](#)" — is intended for students, parents, and educators, in order to spur more interest in our field.

Women will represent 20 percent of the global cybersecurity workforce by the end of 2019.

